

A three-point finite-field Kakeya set

Counterexample to TLMC 00000009118

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Abstract

The three nonzero points in $\mathbb{F}_2^2$ contain an affine line in every nonzero direction. Thus they form a finite-field Kakeya set of cardinality $3 < 2^2$, contradicting the stated lower bound q^n .

1 The exact assertion

The conjecture states that the size of finite-field Kakeya sets has the explicit lower bound q^n . Here q denotes the size of the field and n the ambient dimension. Neither language version supplies a multiplicative constant or excludes fields of small cardinality. We address the asserted coefficient-one bound

$$|K| \geq q^n.$$

This is distinct from a bound $|K| \geq c_n q^n$ with a dimension-dependent constant $c_n < 1$; the example below does not contradict such bounds.

A subset $K \subseteq \mathbb{F}_q^n$ is a Kakeya set if, for every nonzero vector $v \in \mathbb{F}_q^n$, there exists $a \in \mathbb{F}_q^n$ such that the entire affine line

$$\{a + tv : t \in \mathbb{F}_q\}$$

lies in K . For nonzero v , this line has exactly q points.

2 Explicit counterexample

Let $q = 2$, $n = 2$, and work with the field $\mathbb{F}_2 = \{0, 1\}$, with arithmetic modulo two. Define

$$K = \mathbb{F}_2^2 \setminus \{(0, 0)\} = \{(1, 0), (0, 1), (1, 1)\}.$$

Proposition 1. *K is a Kakeya set and $|K| = 3 < q^n = 4$.*

Proof. The following table lists all nonzero directions and a base point in each direction. Because the scalar parameter takes only the values 0 and 1, the two rightmost entries give the complete line.

Direction v	Base point a	$a + 0v$	$a + 1v$
$(1, 0)$	$(0, 1)$	$(0, 1)$	$(1, 1)$
$(0, 1)$	$(1, 0)$	$(1, 0)$	$(1, 1)$
$(1, 1)$	$(1, 0)$	$(1, 0)$	$(0, 1)$

Every listed point belongs to K , and all nonzero directions have been covered. The three points of K are distinct. Hence K is a Kakeya set of size three, whereas $q^n = 2^2 = 4$. $\square$

3 Formal verification

The Lean project uses Lean 4.19.0 and Mathlib commit `c44e0c8ee63ca166450922a373c7409c5d26b00b`. The ambient space `Point` is the actual function space `Fin 2 -> ZMod 2`; Mathlib supplies the field structure on `ZMod 2` and the coordinatewise vector operations. The definition `affinePoint` uses ordinary vector addition and scalar multiplication $a + t \cdot v$.

The predicate `IsKakeya` quantifies over all actual nonzero vectors, existentially quantifies over all actual base points, and requires membership for every scalar. The set `puncturedPlane` is the finite universe with the zero vector erased. The theorem `puncturedPlane_isKakeya` verifies that full quantified predicate; it does not assume a line-incidence table. The additional theorem `affinePoint_injective` verifies that every nonzero direction parameterizes two distinct points.

All finite checks use the kernel-checked `decide` tactic, with no native evaluation, numerical approximation, or external computation. The field size, ambient cardinality, and counterexample cardinality are checked separately. The theorem `conjecture_00000009118` combines the Kakeya property with the strict inequality. The theorem `universal_bound_false` explicitly negates the proposed bound for all Kakeya sets in this ambient space, which suffices to disprove a universal bound over all finite fields and dimensions.

From the `lean` directory, run `lake build`, then `lake env lean Main.lean` for the direct verification and axiom audit. The audited theorems depend only on the standard logical axioms `propext`, `Classical.choice`, and `Quot.sound`. No unproved mathematical assumption is used.